



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences across **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

Rows in the dataset

18

Features

Columns per record

37

Missing Values

In Review Rating column

\$59.76

Avg. Purchase

Average spend per order

Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season), and shopping behavior signals (discounts, promo codes, frequency, review ratings, shipping type).

Data Preparation in Python

Cleaning & Standardization

Loaded data via `pandas`, checked structure with `df.info()` and `.describe()`. Imputed missing Review Ratings using median per category. Renamed all columns to `snake_case`.

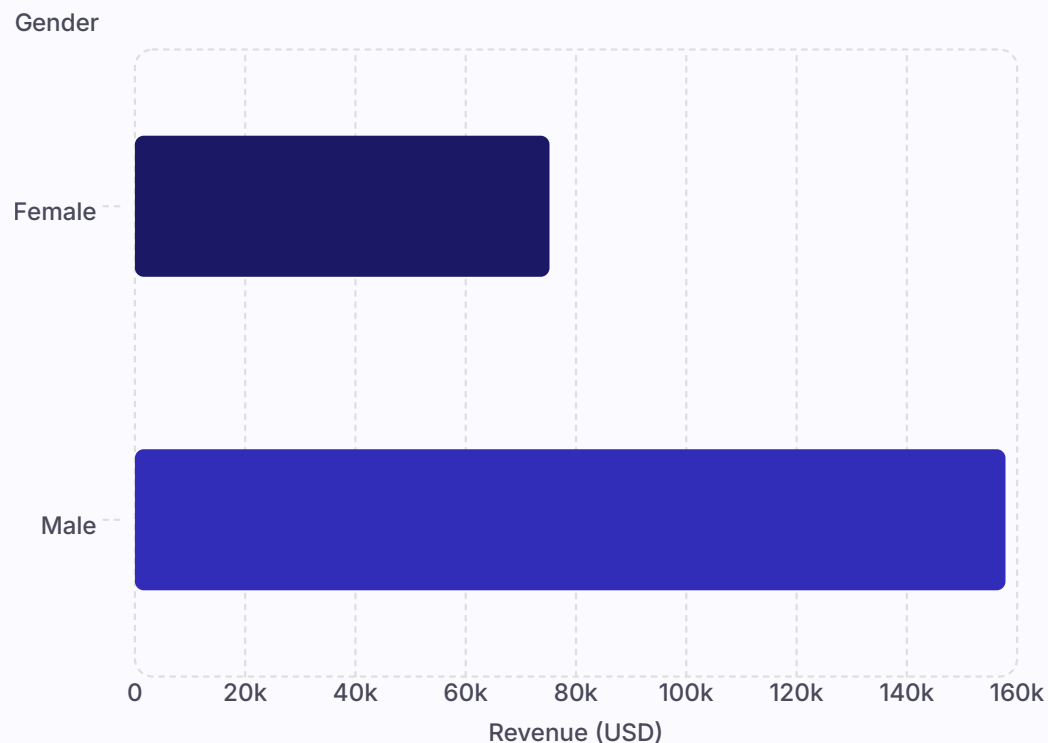
Feature Engineering

Created `age_group` by binning customer ages. Derived `purchase_frequency_days` from purchase data. Verified `discount_applied` and `promo_code_used` were redundant — dropped the latter.

Database Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame for SQL-based analysis.

Revenue Insights: Gender & Shipping



Male Customers Drive 2× Revenue

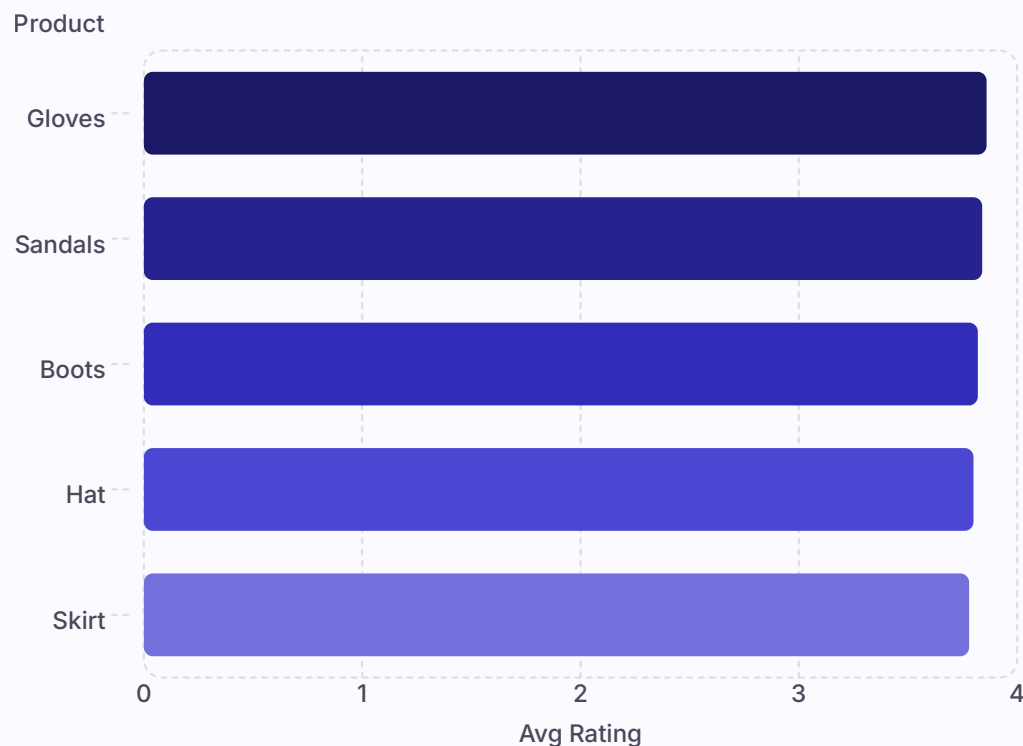
Male customers generated **\$157,890** vs. **\$75,191** for females — a significant spend gap worth investigating.

Shipping Type Impact

Express shipping customers spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**), suggesting higher intent buyers opt for faster delivery.

Top Products & Discount Patterns

Top 5 Products by Rating



Most Discount-Dependent Products

Hat — 50.00%

Highest discount rate of
any product

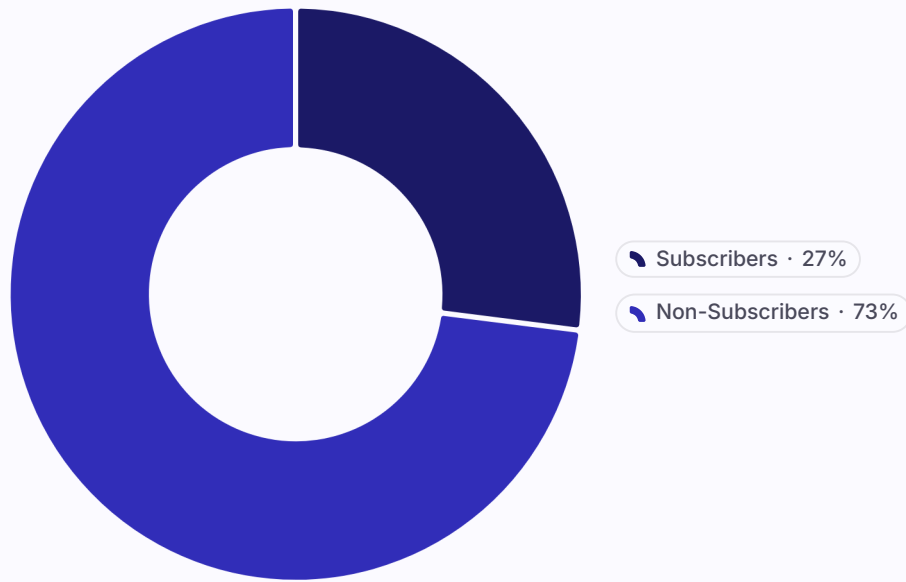
Sneakers — 49.66%

Nearly half of purchases
use discounts

Coat — 49.07%

Strong seasonal discount reliance

Subscription Behavior



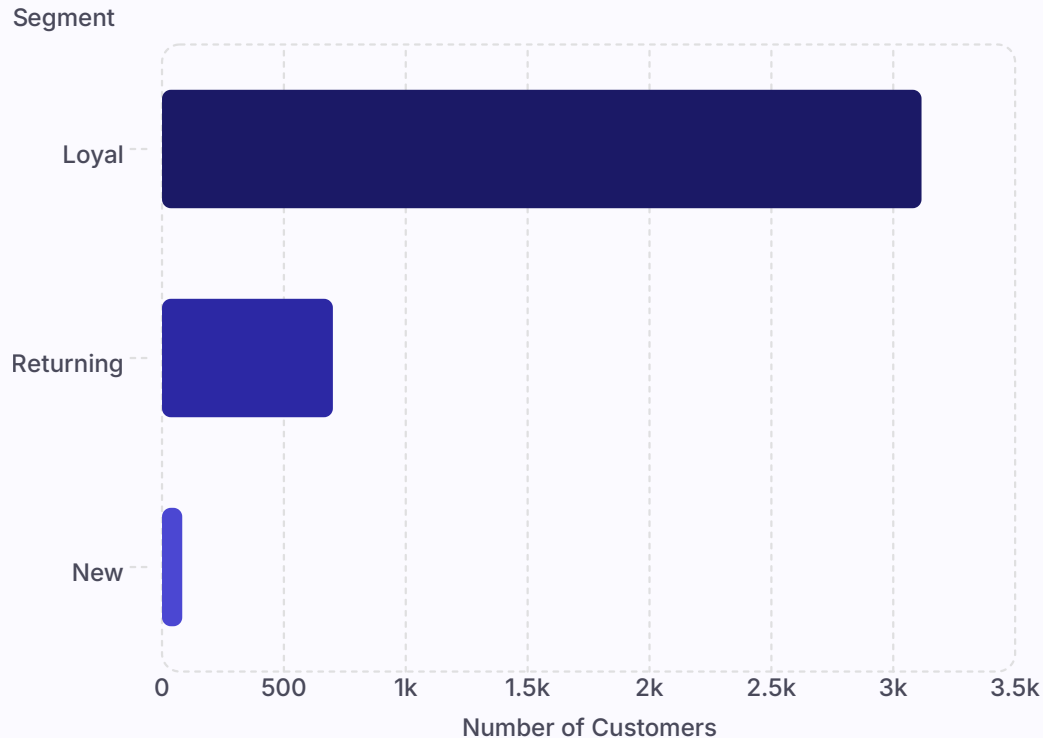
Subscribers vs. Non-Subscribers

Only **27%** of customers subscribe, yet they show similar average spend (~\$59.50). The real opportunity is converting the **73% non-subscribers**.

1,053 Subscribers
\$62,645 total revenue

2,847 Non-Subscribers
\$170,436 total revenue

Customer Segmentation



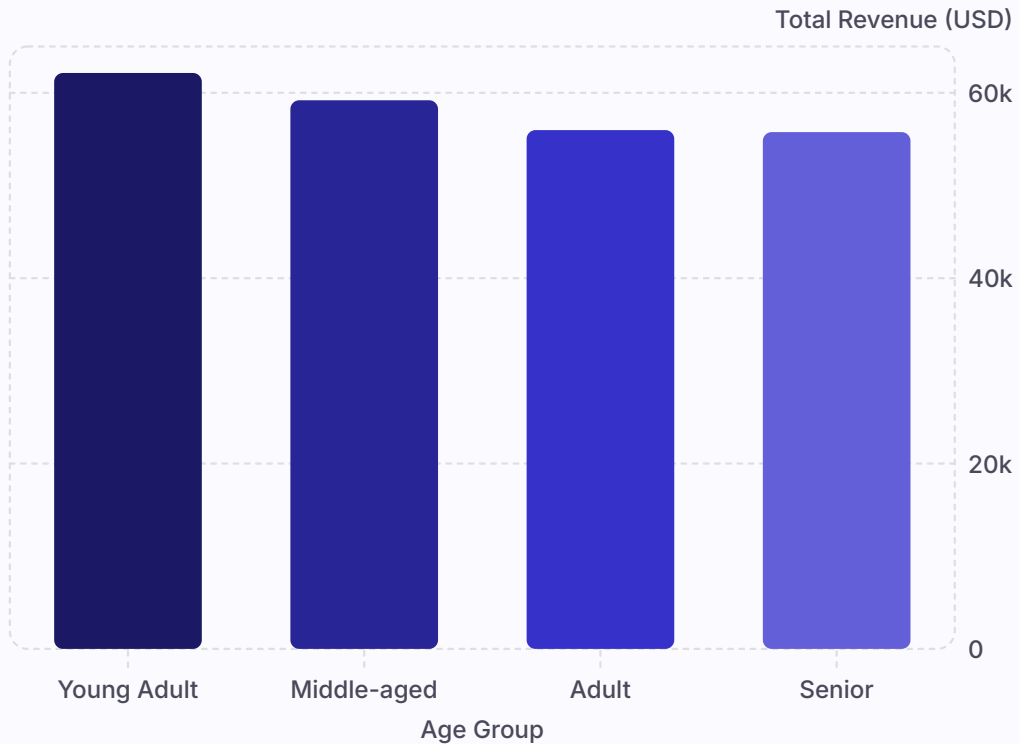
Loyal Customers Dominate

3,116 customers (80%) are classified as Loyal based on purchase history. Only 83 are New customers — acquisition may need a boost.

Repeat Buyers & Subscriptions

Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — suggesting repeat buying alone doesn't drive subscriptions.

Revenue by Age Group



Young Adults Lead Revenue

Young Adults contribute the most revenue at **\$62,143**, with a gradual decline across older groups. Revenue distribution is relatively balanced — all groups contribute meaningfully.

- Average customer age is **44 years** (range 18–70), with 25th percentile at 31 and 75th at 57.

Top Products by Category



Accessories

Jewelry (171 orders),
Sunglasses (161), Belt (161)



Clothing

Blouse (171 orders), Pants
(171), Shirt (169)



Footwear

Sandals (160 orders), Shoes
(150), Sneakers (145)



Outerwear

Jacket (163 orders), Coat (161)

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to move them into the Loyal segment (currently 80% of customers).



Review Discount Policy

Balance sales boosts with margin control — Hats and Sneakers have ~50% discount rates.



Targeted Marketing

Focus campaigns on Young Adults, express-shipping users, and top-rated products like Gloves and Sandals.

